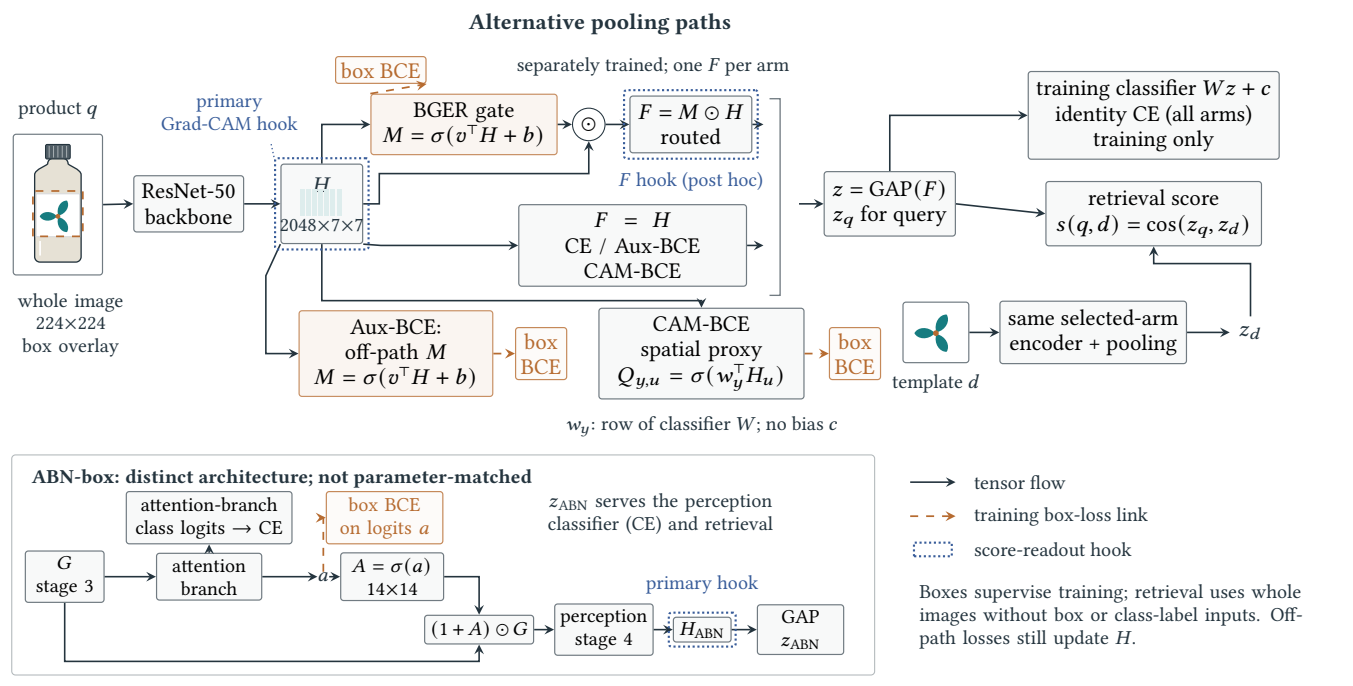

Alternative pooling paths
product q
whole image
224×224
box overlay
primary
Grad-CAM hook
ResNet-50
backbone
H
2048×7×7
box BCE
BGER gate
$M = \sigma(v^\top H + b)$
separately trained; one F per arm
$F = M \odot H$
routed
F hook (post hoc)
$F = H$
CE / Aux-BCE
CAM-BCE
Aux-BCE:
off-path M
$M = \sigma(v^\top H + b)$
box
BCE
CAM-BCE
spatial proxy
$Q_{y,u} = \sigma(w_y^\top H_u)$
box
BCE
$z = \mathrm{GAP}(F)$
z_q for query
training classifier $Wz + c$
identity CE (all arms)
training only
retrieval score
$s(q, d) = \cos(z_q, z_d)$
template d
same selected-arm
encoder + pooling
z_d
w_y: row of classifier W; no bias c
ABN-box: distinct architecture; not parameter-matched
G
stage 3
attention branch
attention-branch
class logits → CE
box BCE
on logits a
a
$A = \sigma(a)$
14×14
$(1 + A) \odot G$
primary hook
perception
stage 4
H_{ABN}
GAP
z_{ABN}
z_{ABN} serves the perception
classifier (CE) and retrieval
tensor flow
training box-loss link
score-readout hook
Boxes supervise training; retrieval uses whole images without box or class-label inputs. Off-path losses still update H.